Summary of CSS3 Advanced Properties

CSS Opacity / Transparency

The `opacity` property specifies the opacity/transparency of an element.

Transparent Image

The `opacity` property can take a value from 0.0 - 1.0. The lower value, the more transparent:



opacity 0.2 | opacity 0.5 | opacity 1 (default)

```
img {
    opacity: 0.5;
    filter: alpha(opacity=50); /* For IE8 and earlier */
}
```

Transparent Hover Effect

The `opacity` property is often used together with the `:hover` selector to change the opacity on mouse-over:

```
img {
    opacity: 0.5;
    filter: alpha(opacity=50); /* For IE8 and earlier */
}

img:hover {
    opacity: 1.0;
    filter: alpha(opacity=100); /* For IE8 and earlier */
}
```

CSS filter Property

`filter` property specifies the effect to apply to a image.

Example: Change all images to black and white (100% gray):

```
img {
   filter: grayscale(100%);
}
```

`filter` toma como valor una función que hace un retoque a la imagen:

```
filter: opacity(25%);
filter: saturate(30%);
filter: grayscale(50%);
filter: sepia(60%);
filter: blur(5px);
filter: brightness(0.4);
filter: contrast(200%);
filter: drop-shadow(16px 16px 20px blue);
filter: invert(75%);
```

Podéis probarlas aquí:
https://www.w3schools.com/cssref/playdemo.php?filename=playcss_filter

Se puede aplicar más de una función a la propiedad filter:

```
filter: contrast(175%) brightness(3%);

filter: drop-shadow(3px 3px red) sepia(100%) drop-shadow(-3px -3px blue);
```

CSS3 border-image Property

The CSS3 `border-image` property allows you to specify an image to be used instead of the normal border around an element.

The property has three parts:

1. The image to use as the border
2. Where to slice the image
3. Define whether the middle sections should be repeated or stretched

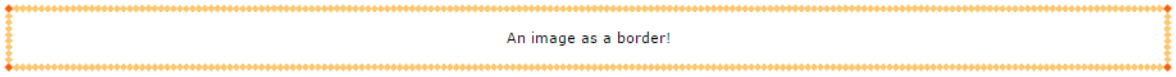
We will use the following image (called "border.png"):



The `border-image` property takes the image and slices it into nine sections, like a tic-tac-toe board. It then places the corners at the corners, and the middle sections are repeated or stretched as you specify.

Note: For `border-image` to work, the element also needs the `border` property set!

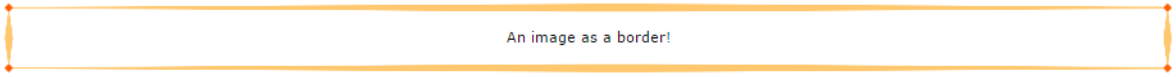
Here, the middle sections of the image are repeated to create the border:



Here is the code:

```
#borderimg {
    border: 10px solid transparent;
    padding: 15px;
    -webkit-border-image: url(border.png) 30 round; /* Safari 3.1-5 */
    -o-border-image: url(border.png) 30 round; /* Opera 11-12.1 */
    border-image: url(border.png) 30 round;
}
```

Here, the middle sections of the image are stretched to create the border:



Here is the code:

```
#borderimg {
    border: 10px solid transparent;
    padding: 15px;
    -webkit-border-image: url(border.png) 30 stretch; /* Safari 3.1-5 */
    -o-border-image: url(border.png) 30 stretch; /* Opera 11-12.1 */
    border-image: url(border.png) 30 stretch;
}
```

Try it Yourself »

Tip: The `border-image` property is actually a shorthand property for the `border-image-source`, `border-image-slice`, `border-image-width`,`border-image-outset` and `border-image-repeat` properties.

CSS3 Backgrounds

CSS3 contains a few new background properties, which allow greater control of the background element.

In this chapter you will learn how to add multiple background images to one element.

You will also learn about the following new CSS3 properties:

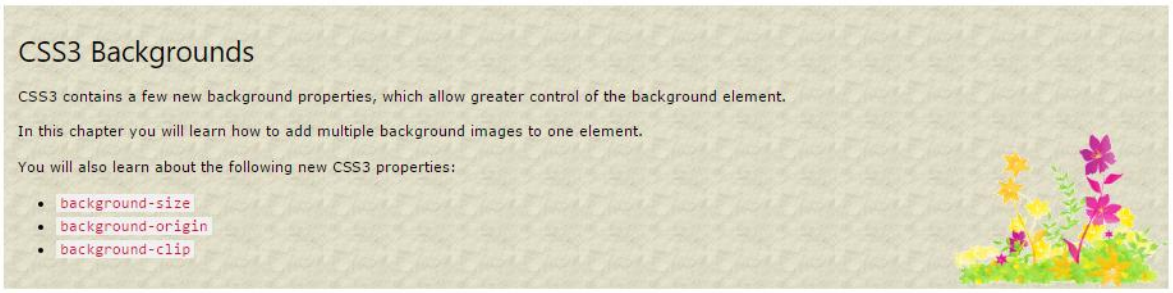
- `background-size`
- `background-origin`
- `background-clip`

CSS3 Multiple Backgrounds

CSS3 allows you to add multiple background images for an element, through the `background-image` property.

The different background images are separated by commas, and the images are stacked on top of each other, where the first image is closest to the viewer.

The following example has two background images, the first image is a flower (aligned to the bottom and right) and the second image is a paper background (aligned to the top-left corner):




```
#example1
{
    background-image: url(img_flwr.gif), url(paper.gif);
    background-position: right bottom, left top;
    background-repeat: no-repeat, repeat;
}
```

Multiple background images can be specified using either the individual background properties (as above) or the `background` ***shorthand*** property:

```
#example1 {
    background: url(img_flwr.gif) right bottom no-repeat, url(paper.gif) left
top repeat;
}
```

CSS3 Background Size

The CSS3 `background-size` property allows you to specify the size of background images.

Before CSS3, the size of a background image was the actual size of the image. CSS3 allows us to re-use background images in different contexts.

The size can be specified in lengths, percentages, or by using one of the two keywords: contain or cover.

The following example resizes a background image to much smaller than the original image (using pixels):



Here is the code:

```
#div1 {
    background: url(img_flower.jpg);
    background-size: 100px 80px;
    background-repeat: no-repeat;
}
```

The two other possible values for `background-size` are `contain` and `cover`.

The `contain` keyword scales the background image to be as large as possible (but both its width and its height must fit inside the content area). As such, depending on the proportions of the background image and the background positioning area, there may be some areas of the background which are not covered by the background image.

The `cover` keyword scales the background image so that the content area is completely covered by the background image (both its width and height are equal to or exceed the content area). As such, some parts of the background image may not be visible in the background positioning area.

The following example illustrates the use of `contain` and `cover`:

```
#div1 {
    background: url(img_flower.jpg);
    background-size: contain;
    background-repeat: no-repeat;
}
#div2 {
    background: url(img_flower.jpg);
    background-size: cover;
    background-repeat: no-repeat;
}
```

+ Propiedades relacionadas:

http://www.w3schools.com/cssref/css3_pr_background-clip.asp
http://www.w3schools.com/cssref/css3_pr_background-origin.asp